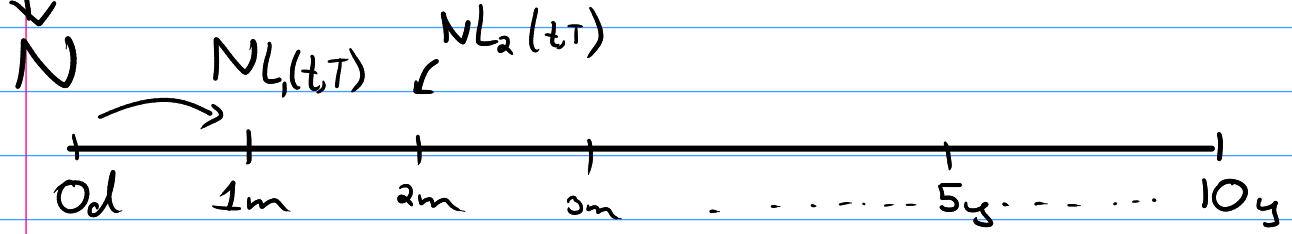
→ Caps: Cobertura ante el alza en las tasas de interés

100,000,000

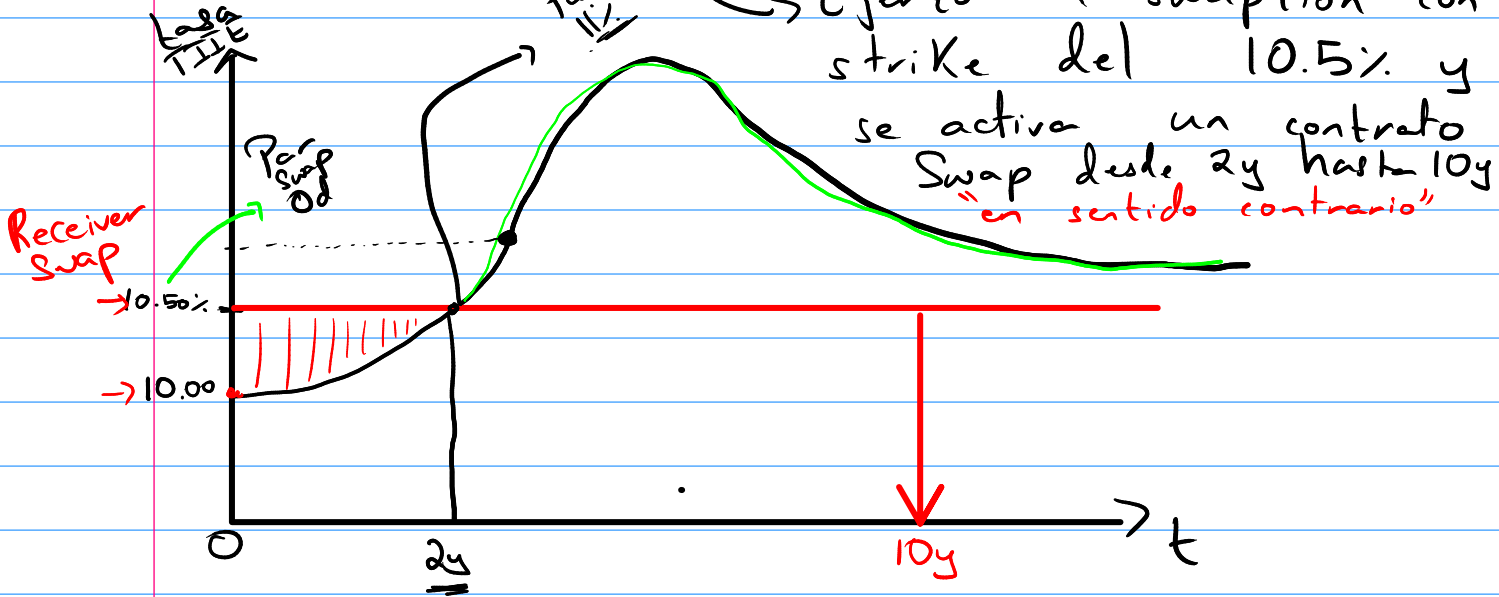


→ IRS ⇝ Swap ($L_1^{Float}(t,T) \leftrightarrow$ Fija)

→ CAP ⇝ cap (Pagando Fija ; si $L_1(t,T) >$ Fija / $L_1(t,T)$ si $L_1(t,T) <$ Fija)

↳ $(L_1 - K)$

Swaption



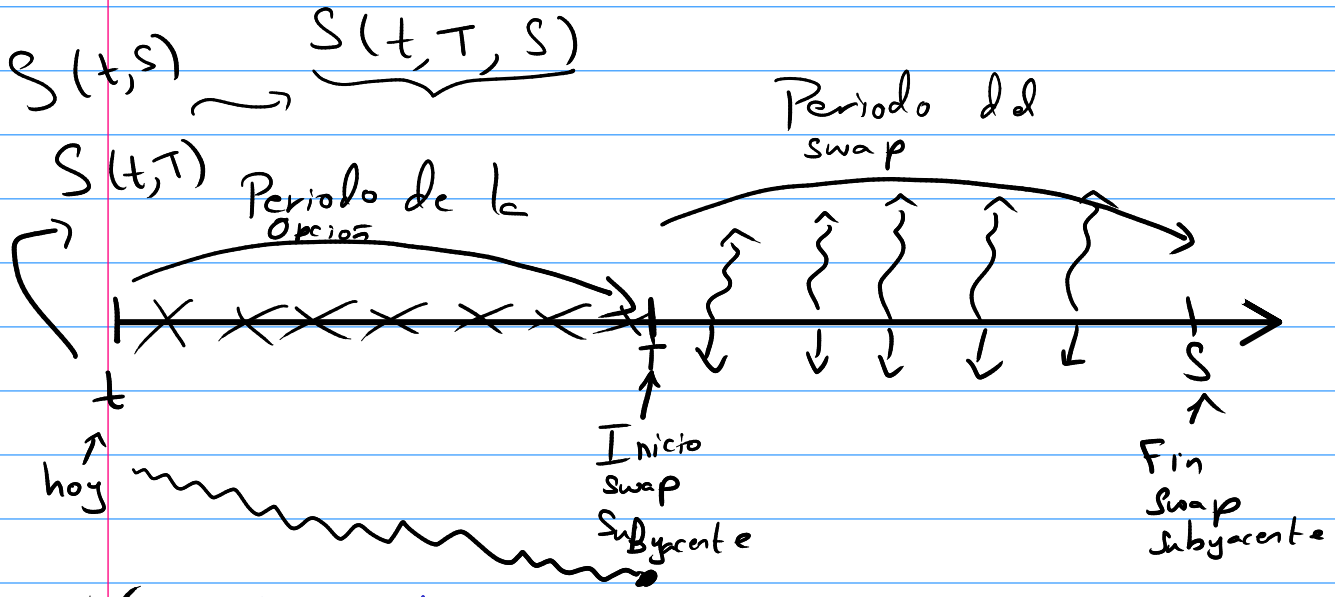
→ Ejerzo el swaption con strike del 10.5% y se activa un contrato Swap desde 2y hasta 10y "en sentido contrario"

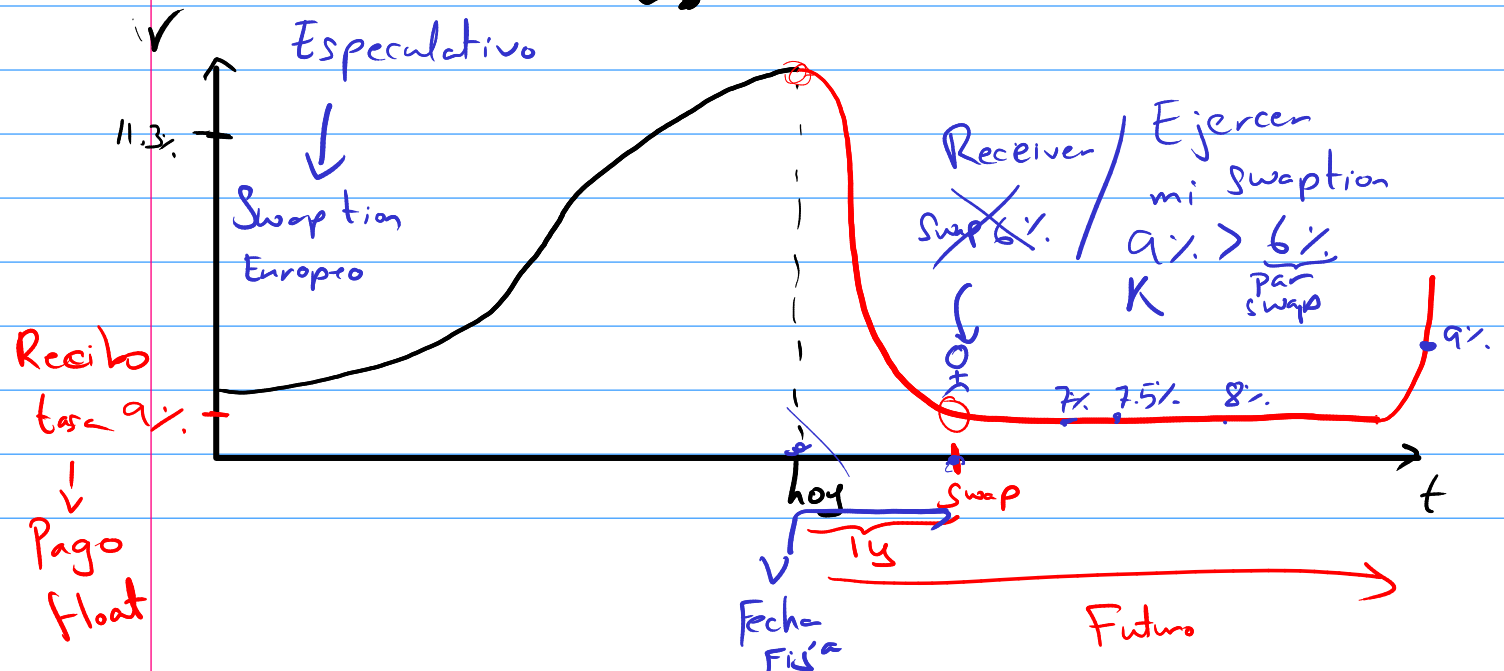
Instrumento / Opción de entrar en un swap

Swap + Swaption } Swap cancelable

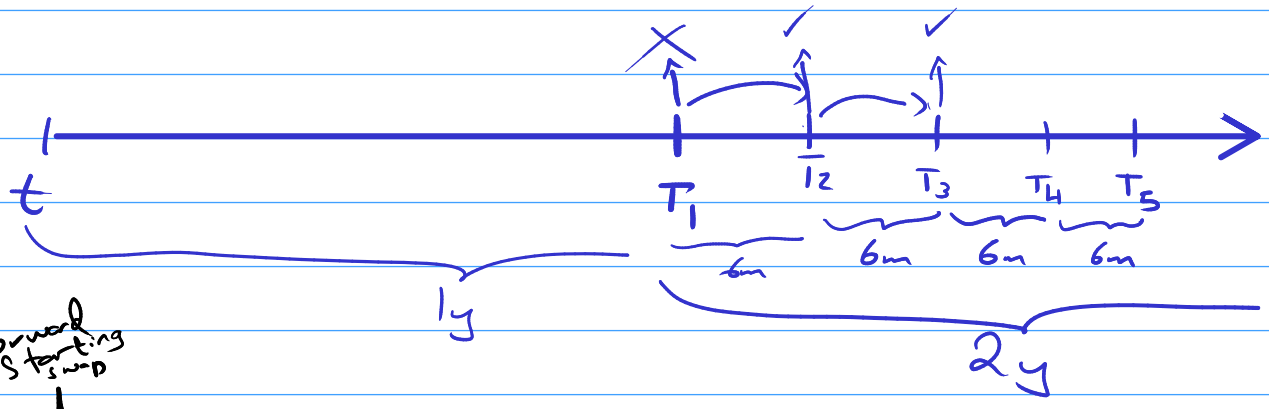
N

	Recibo Fix	Paso Float	Recibe Float	Pago Fija
	10.3%			
✓ 1y	~~10.5%~~	10%	—	—
✓ 2y	10.3% ~~10.5%~~	10.3%	—	—
✗ 3y	~~10.5%~~	10.6%	10.6%	10.5%
✗ 4y	10.5%	10.9%	10.9	10.5
✗ 5y	10.5%	11.0%	11	10.5
✗ 6y	10.5%	11.2%	11.2	10.5
✗ 7y	10.5%	11.1%	11.1	10.5
✗ 8y	10.5%	11.05%	11.05	10.5
✗ 9y	10.5%	10.98%	10.98	10.5
✗ 10y	10.5%	10.75%	10.75	10.5





Swaption Bermuda

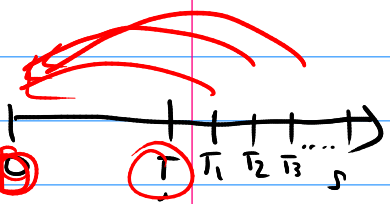


Forward Starting Swap

Hoy (t) Swap empiece en T y acabe en S

$$= \text{Pata Fija} - \text{Pata Float}$$

$$= \sum_{i=1}^{n} \tau(T_{i-1}, T_i) P(0, T_i) S(0, T, S)$$



$$- \sum_{i=1}^{n} \tau(T_{i-1}, T_i) P(0, T_i) F(0, T_{i-1}, T_i)$$

en T Pactamos un swap

$$= \sum_{i=1}^{n} \tau(T_{i-1}, T_i) P(T, T_i) S(T, T, S)$$

$$- \sum_{i=1}^{n} \tau(T_{i-1}, T_i) P(0, T_i) F(0, T_{i-1}, T_i)$$

$$\text{Payoff} = \sum_{i=1}^{n} \tau(T_{i-1}, T_i) P(T, T_i) \left[S_{Swap}(T, T, S) - K \right]$$

Payoff

Receiver swaption

Swaption

$$\sum_{i=1}^{n} \tau(T_{i-1}, T_i)\, P(T, T_i)\, \max\left(S_{swap}(T, T, s) - K, 0\right)$$

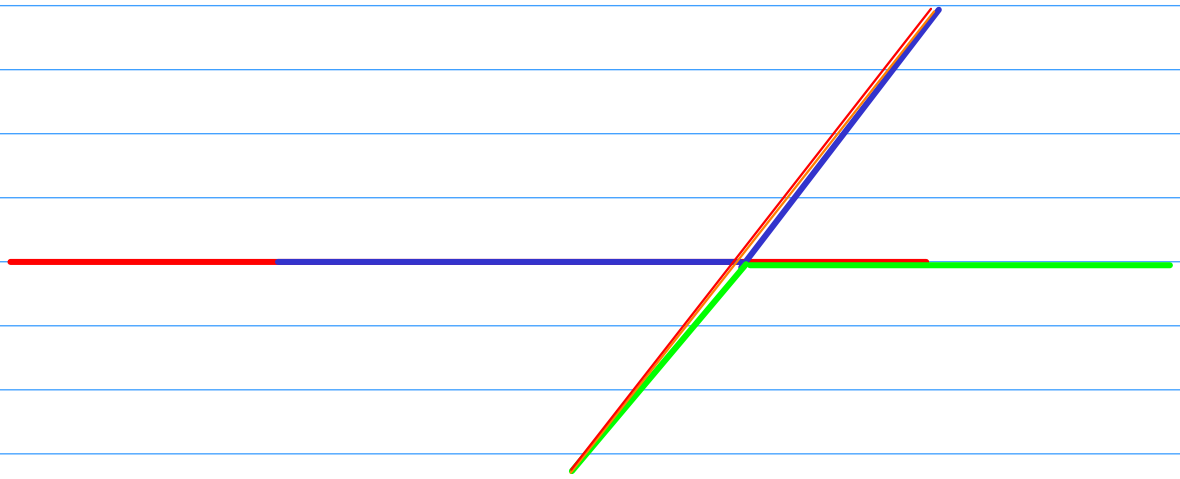
$$= \max\left(S_{swap}(T, T, s) - K\right) \underbrace{\sum_{i=1}^{n} \tau(T_{i-1}, T_i)\, P(T, T_i)}_{A_n}$$

$$\boxed{= A_n \cdot \max\left(S_{swap}(T, T, s) - K\right)}$$

$$S = \frac{\sum_{i=1}^{n} \tau_i P(t_i) F(t_i)}{\sum_{i=1}^{n} \tau_i P(t_i)} \quad \text{"Promedio"}$$



Payer Swaption − Receiver Swaption = Fwd Starting Swap



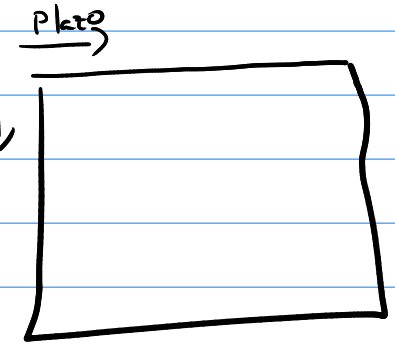
Problemas en la valoración de opciones.

→ No conocemos "a priori" la volatidad utilizada como insumo para valorar opciones,

* Tenemos que recurrir a calcular una Volatilidad Implícita.

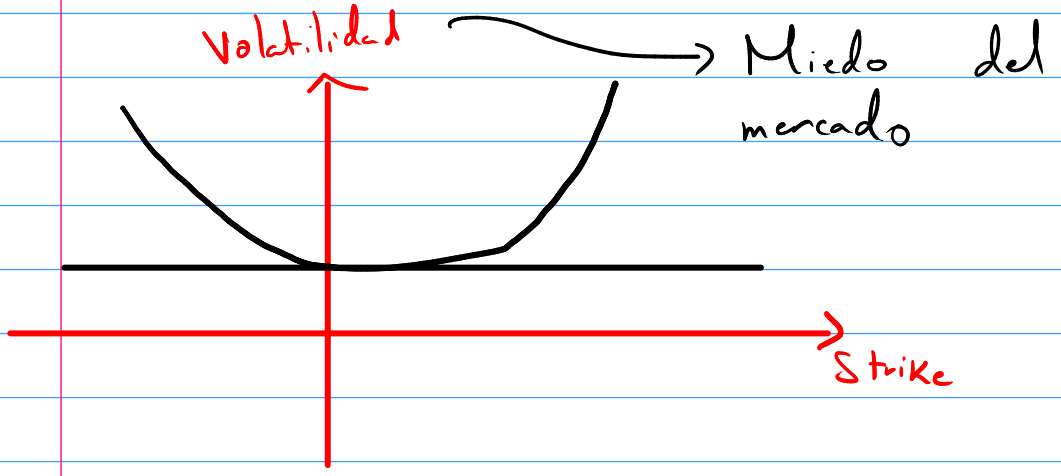
FX Options, venc. 1y, USDMXN

	BID	MID	OFFER
K = 16.5	$ 20	→ 20.5 ←	$ 21
K = 17	$ 19	19.5	$ 20
K = 17.5	$ 18	18.5	$ 19
K = 18	$ 17	17.5	$ 18



Volatilidad Implícita:

→ El valor de la volatilidad, tal que introduciéndola en la fórmula de Black & Scholes / Black 76 nos regrese el precio de la opción observado en pantalla.

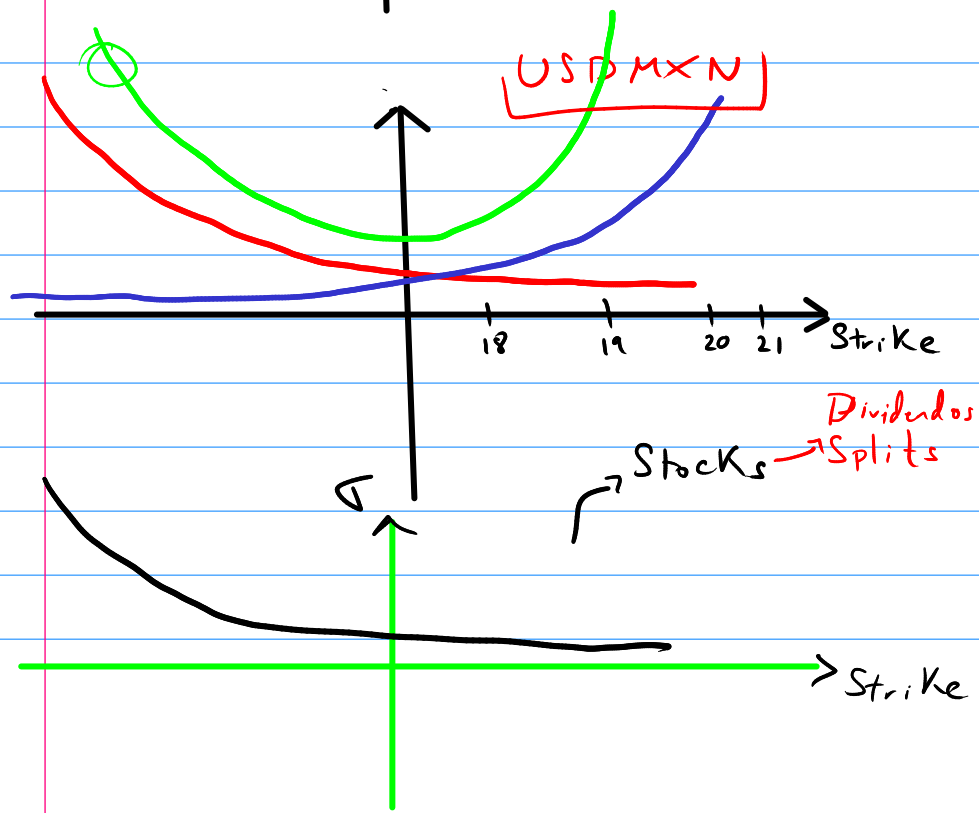


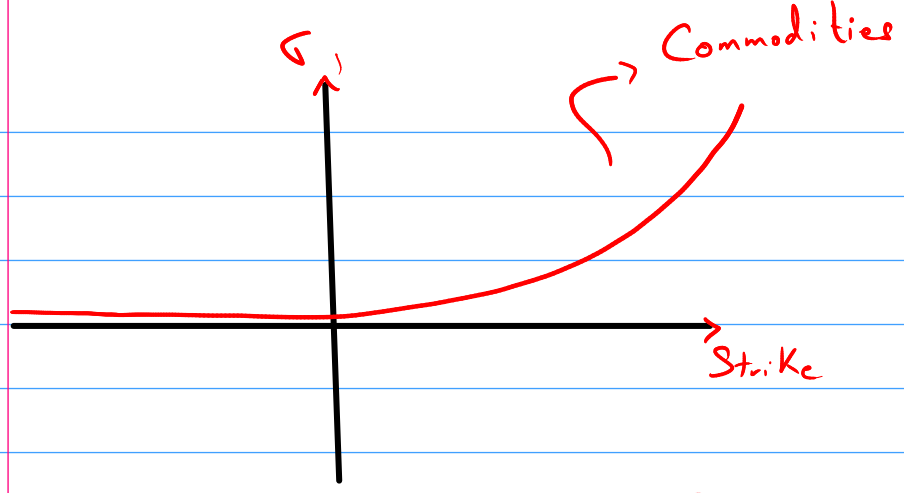
→ Market Makers / Formadores de Mercado

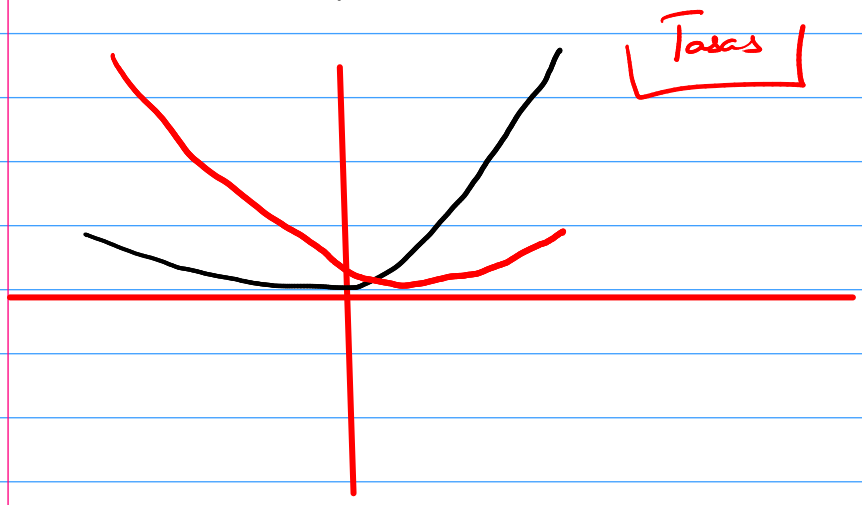
* Proveer de liquidez en los mercados financieros

1y USDMXN ~ $20

	Bids	Offers/Ask
K = 18.5	$50	$51
K = 20	$53	$55




Commodities
Strike


Tasas